Hainan GeoInsight | Sanya

REAL OpenStreetMap data | Walking speed 4.8 km/h | EPSG:32649

source: User local hainan-latest.osm.pbf; administrative boundaries from OSM Nominatim

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study_area_km2: 73.32984036287378

scope: OSM administrative polygon intersect mapped developed land/buildings and 300m walking-road proximity; subtract mapped water/forest/agriculture/natural undeveloped classes

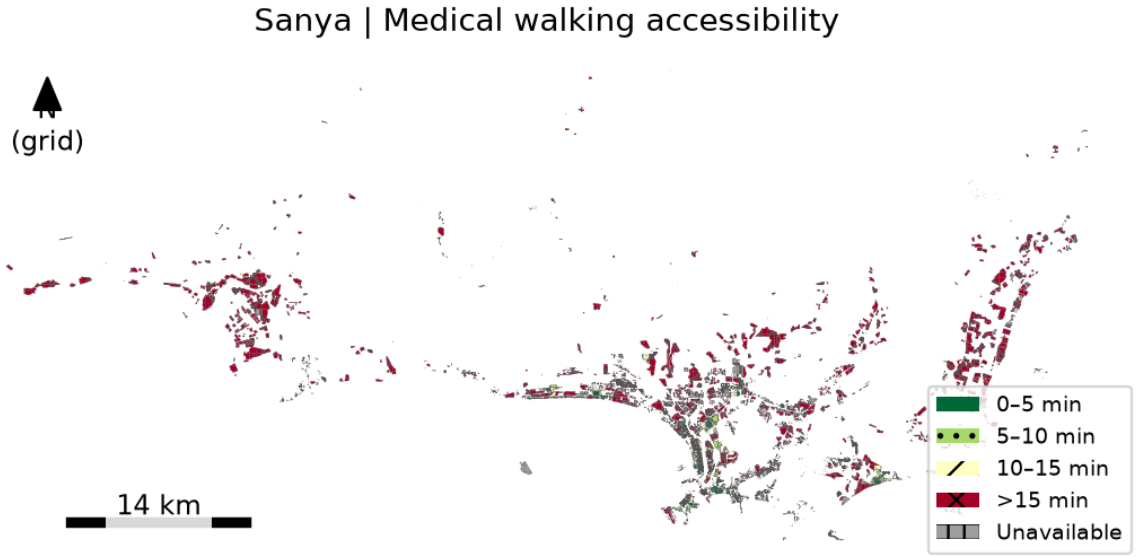
Methods: administrative boundary intersected with mapped developed polygons and 300 m proximity to walking roads, minus mapped water, forest and undeveloped classes. Each clipped grid receives one interior representative sampling point. Directed multi-source Dijkstra includes straight endpoint offsets, maximum snapping distance 300 m. Coverage denominator: all included grid samples, not population.

Facility	Grid samples	Reachable	Median min	15-min coverage
medical	1618	1499	90.15	7.0%
education	1618	1506	46.75	11.8%
transit	1618	1500	64.75	6.9%

Data Limitations: OSM data may have missing facilities. The grid analysis uses uniform sampling points, not actual population weights. Walking speed is set as a constant (4.8 km/h), ignoring terrain and age differences.

Conservative mapped-developed footprint, not complete city land cover. Unmapped developed areas omitted; OSM classification errors possible. Network is full administrative-city walking graph; no outside-city paths. Relation POIs omitted.

Medical facilities per grid is an unweighted cell mean, never a per-capita measure. Grid north differs slightly from true north due to UTM convergence. Routes remain within city limits; barriers and facility entrances are not fully modeled.



Discrete thresholds and distinct hatches encode travel time independently of color. Grey cross-hatch indicates unavailable routes; it is not zero travel time.